Substitution ciphers

Simple Substitution Cipher

- The message is encrypted by substituting the letter of the alphabet *n* places ahead of the current letter.
- For example, with *n* = 3, the substitution—which acts as the key is:

Plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

where we've followed the convention that the plaintext is lowercase and the ciphertext is uppercase.

Simple Substitution

- Plaintext: fourscoreandsevenyearsago
- Key:

Plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

- Ciphertext:
  IRXUVFRUHDQGVHYHQBHDUVDJR
- Shift by 3 is “Caesar’s cipher”

Ceasar's Cipher

- For each plaintext letter p, substitute the ciphertext letter C:
- $C = E(3, p) = (p + 3) \bmod 26$
- A shift may be of any amount, so that the general Caesar algorithm is
- $C = E(k, p) = (p + k) \bmod 26$

Ceasar's Cipher Decryption

- Suppose we know a Caesar's cipher is being used:

Plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

- Given ciphertext:

  VSRQJHEREVTXDUHSDWV

  - Plaintext: spongebobsquarepants

Cryptanalysis

- If we limit the simple substitution to shifts, then the possible keys are $n \in \{0, 1, 2, \ldots, 25\}$.
- A simple substitution (shift by n) is used
  - But the key is unknown
- Given ciphertext: CSYEVIXIVQMREXIH
- How to find the key?
- Only 26 possible keys — try them all!

Cryptanalysis

- Decrypting the message with each putative key and checking whether the resulting putative plaintext looks like sensible plaintext.
- The brute force approach of trying all possible keys until we stumble across the correct one is known as an *exhaustive key search*.
- It's necessary (although far from sufficient) that the number of possible keys be too large to simply try them all in any reasonable amount of time.

Simple Substitution: General Case

- The simple substitution cipher need not be limited to shifting by *n*.
- In general, simple substitution key can be any **permutation** of letters
  - Not necessarily a shift of the alphabet
- For example

Plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext	J	I	C	A	X	S	E	Y	V	D	K	W	B	Q	T	Z	R	H	F	M	P	N	U	L	G	O

  - Then $26! > 2^{88}$ possible keys

Playfair Cipher

- The best-known multiple-letter encryption cipher is the Playfair.
- The Playfair algorithm is based on the use of a 5 x 5 matrix of letters constructed using a keyword.
- If the keyword is ***monarchy***. The matrix is constructed by filling in the letters of the keyword (minus duplicates) from left to right and from top to bottom, and then filling in the remainder of the matrix with the remaining letters in alphabetic order. The letters I and J count as one letter.

Playfair Cipher

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Playfair Cipher

- Plaintext is encrypted two letters at a time, according to the following rules:

1. Repeating plaintext letters that are in the same pair are separated with a filler letter, such as x, so that balloon would be treated as ba lx lo on.
2. Two plaintext letters that fall in the same row of the matrix Shift Right. For example, ar is encrypted as RM.

Playfair Cipher

3. Two plaintext letters that fall in the same column Shift Down. For example, mu is encrypted as CM.
4. Otherwise, each plaintext letter in a pair is replaced by the letter that lies in its own row and the column occupied by the other plaintext letter. Thus, hs becomes BP and ea becomes IM

Playfair Cipher

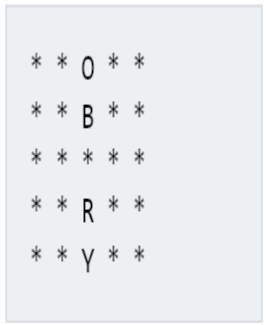
Assume one wants to encrypt the digram OR. There are five general cases:

1)



Hence, OR → YZ

2)



Hence, OR → BY

3)



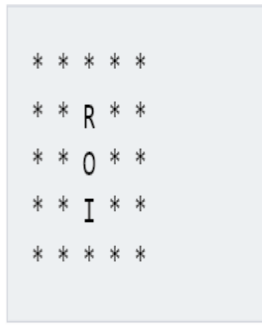
Hence, OR → ZX

4)



Hence, OR → RW

5)



Hence, OR → IO

Playfair Cipher: Example Encryption

- Plaintext: Balloon
- Key: monarchy
- Encryption

BA: IB

LX: SU

LO: PM

ON: NA

Ciphertext: IBSUPMNA

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Playfair Cipher: Example Decryption

- Ciphertext: IBSUPMNA
- Decryption
- Same row: shift Left
- Same column: shift up
- Difference: intersection
- IB: BA
- SU: LX
- PM: LO
- NA: ON

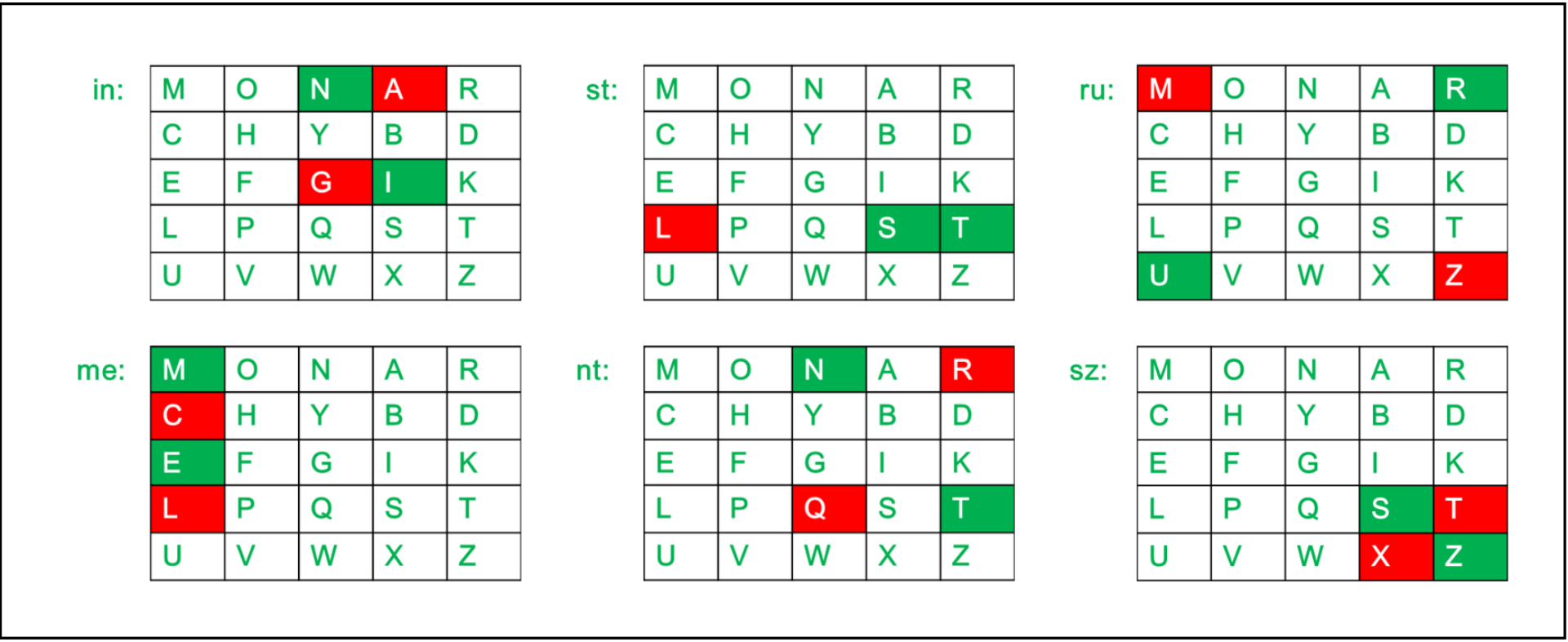
M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Playfair Cipher: Example 2 Encryption

Key: monarchy
Plaintext: instruments

- The plaintext is split into pairs of two letters (digraphs). If there is an odd number of letters, a Z is added to the last letter.
- **PlainText:** `"instruments"`
- **After Split:** `'in' 'st' 'ru' 'me' 'nt' 'sz'`

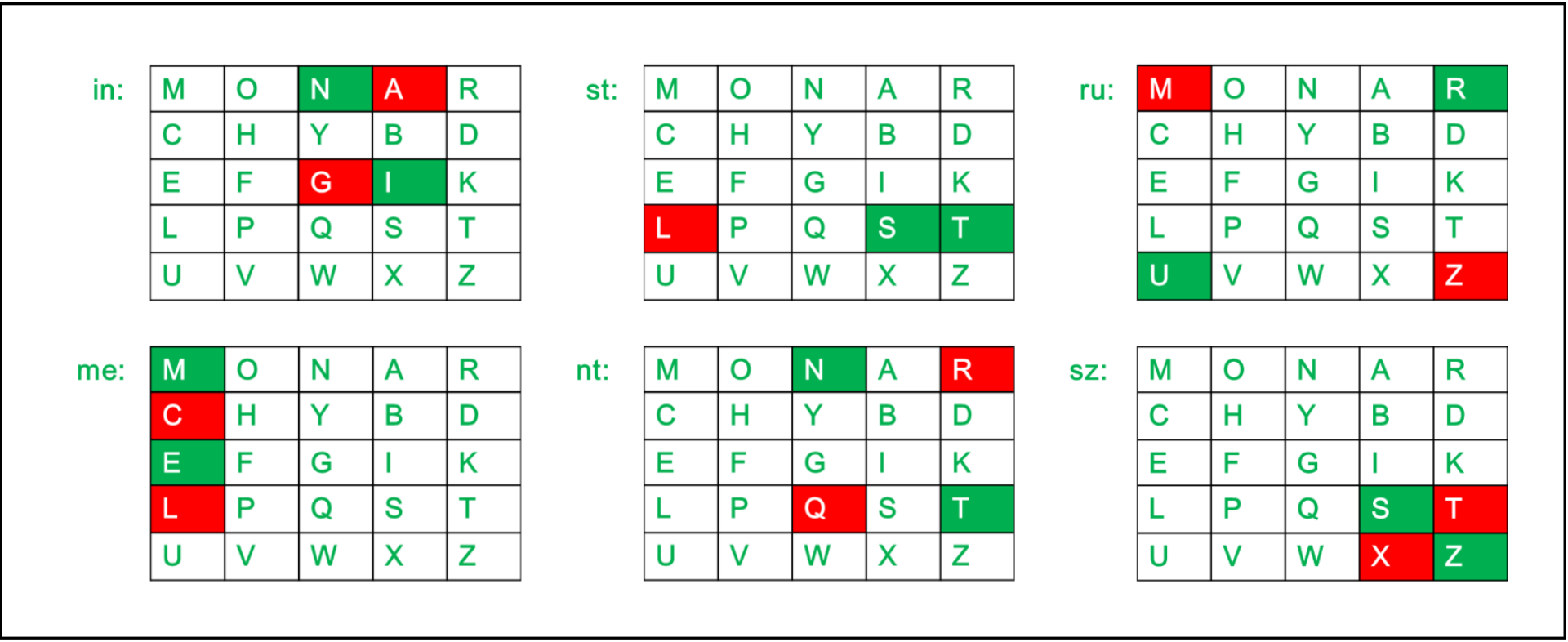
Playfair Cipher: Example 2 Encryption



Playfair Cipher: Example 2 Encryption

- Plain Text: "instrumentsz"
- Encrypted Text: gatlmzclrqtx
- Encryption:
- i -> g
- n -> a
- s -> t
- t -> l
- r -> m
- u -> z
- m -> c
- e -> l
- n -> r
- t -> q
- s -> t
- z -> x

Playfair Cipher: Example 2 Decryption



Playfair Cipher: Example 2 Decryption

- Plain Text: "gatlmzclrqtx"
- Decrypted Text: instrumentsz
- Decryption:
- (red)-> (green)
- ga -> in
- tl -> st
- mz -> ru
- cl -> me
- rq -> nt
- tx -> sz

PROBLEM

- Using "playfair example" as the key, encrypt the message "hide the gold in the tree stump" using playfair cipher.

Playfair Cipher Drawback

- The main drawback of the Playfair cipher is that:
- The plaintext can consist of 25 uppercase letters only. One letter has to be omitted and cannot be reconstructed after decryption.
- Also lowercase letters, white space, numbers and other printable characters cannot be handled by the traditional cipher. This means that complete sentences cannot be handled by this cipher.
- Space between two words in the plaintext is not considered as one character.
- A spare letter X is added when the plaintext word consists of odd number of character. In the decryption process this X is ignored. X is a valid character and creates confusion because it could be a part of plaintext, so we cannot simply remove X in decryption process.